

Do Structured Probabilistic Models Still Matter for Low-Resource Bengali Named Entity Recognition?

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Abstract—Named Entity Recognition (NER) for low-resource languages remains challenging because modern neural architectures require large annotated corpora that are unavailable for most languages. Bengali, the seventh most spoken language with over 230 million native speakers, is still severely under-resourced for NER tasks. This paper investigates whether structured probabilistic models, particularly Conditional Random Fields (CRFs), can remain competitive with deep neural approaches under extreme data scarcity. We propose a Latent Morphology CRF that models morphological segmentation as a latent variable and jointly learns Bengali suffix distributions with NER through Expectation-Maximization (EM), without requiring explicit morphological annotations. We evaluate four model families—Basic CRF, Latent Morph CRF, BiLSTM, and multilingual BERT (mBERT)—across six training subsets ranging from 1% to 100% of the WikiANN Bengali corpus. Experimental results show that CRF-based models outperform neural baselines in extremely low-resource settings. At 1% training data, the BiLSTM model collapses entirely ($F1 = 0.000$), whereas the Latent Morph CRF achieves an F1-score of 0.509. A performance crossover occurs at approximately 10% training data, beyond which mBERT becomes dominant, achieving an F1-score of 0.947 using the full dataset. An ablation study further reveals that contextual features contribute most significantly to performance ($F1$ drop = 0.032), while morphological features provide modest but consistent improvements. These findings establish practical data-threshold guidelines for developing Bengali NER systems under resource-constrained conditions.

Index Terms—Named Entity Recognition, Conditional Random Fields, Bengali NLP, Low-Resource NLP, Latent Variable Models, Expectation-Maximization, Morphological Segmentation, mBERT

I. INTRODUCTION

Named Entity Recognition (NER) — the task of identifying and classifying mentions of persons, organizations, and locations in text — is a foundational component of information extraction, question answering, and knowledge graph construction pipelines. While state-of-the-art performance on English NER has long been dominated by transformer-based architectures fine-tuned on large annotated corpora [1], the practical deployment of such systems for the majority of the world’s languages remains hindered by a fundamental data bottleneck. Bengali exemplifies this tension. As the seventh most spoken language globally, with an estimated 230 million native speakers concentrated in Bangladesh and the Indian

state of West Bengal, Bengali has significant societal importance for NLP applications ranging from misinformation detection to public health communication. Yet Bengali NER benchmarks remain sparse, annotation resources are limited, and the language’s rich agglutinative morphology introduces complexities that models trained on Indo-European languages do not automatically transfer [2]. The practical question facing Bengali NLP practitioners is therefore not which model achieves the highest benchmark score given unlimited data, but rather: which model is most reliable when only a small fraction of annotated data is available? This paper addresses that question directly. Probabilistic graphical models — in particular, Conditional Random Fields (CRFs) [3] — offer a principled alternative to data-hungry neural approaches. CRFs encode linguistic knowledge directly through hand-crafted feature functions: morphological patterns, character n-grams, and contextual word features. These explicit inductive biases can substitute for the statistical signal that neural models must learn from large corpora. Under extreme data scarcity, this distinction may be decisive. We extend the standard CRF baseline with a novel Latent Morphology CRF that treats the morphological segmentation of each Bengali token as a latent variable z . Inspired by latent variable CRF frameworks [4, 5], the model marginalizes over all possible suffix decompositions of each token and learns a prior distribution over suffix lengths via EM. Crucially, this requires no morphological lexicon or external segmenter — the morphological structure is discovered jointly with the NER task. The contributions of this paper are:

The first systematic low-resource benchmark for Bengali NER evaluating four model families across six controlled training data sizes (1%, 5%, 10%, 30%, 50%, 100%). A Latent Morphology CRF that learns Bengali suffix distributions unsupervised via EM, requiring no morphological annotation. An ablation study quantifying the contribution of four feature groups — morphology, character n-grams, context, and word identity — to CRF performance in the low-resource Bengali setting. Empirical identification of the data crossover point (10%) at which structured models lose their advantage over pretrained transformers, providing actionable guidance for practitioners.

The remainder of this paper is organized as follows. Section II reviews related work. Section III formalizes the model. Sec-

tion IV describes the experimental setup. Section V presents results and ablation findings. Section VI discusses implications. Section VII concludes.

II. RELATED WORK

A. Conditional Random Fields for Sequence Labeling

Lafferty et al. [3] introduced CRFs as discriminative undirected graphical models for sequence labeling, addressing the label-bias problem inherent in Maximum Entropy Markov Models (MEMMs). The key insight is global normalization: rather than computing local label probabilities at each position independently, the CRF defines a single joint distribution over the entire label sequence conditioned on the observation sequence. For NER, this means the model considers all possible label sequences simultaneously during training and inference, enabling it to enforce global consistency — such as ensuring I-PER never follows B-ORG — without hard-coded constraints.

B. Latent Variable and Morphology-Aware Models

Quattoni et al. [4] extended CRFs to hidden-state CRFs, introducing latent variables to capture unobserved intermediate representations. Peng and McCallum [5] applied this paradigm to information extraction, demonstrating that latent variables can model ambiguous features without explicit annotation. Our Latent Morph CRF draws directly from this lineage, treating the suffix length of each Bengali token as a discrete latent variable and learning its prior via EM [6] jointly with the NER objective.

C. Bengali NLP and NER

Bengali NER has received growing but still limited attention. The WikiANN benchmark [7] provides a multilingual NER dataset derived from Wikipedia, covering 282 languages including Bengali, with standardized BIO annotation for persons, organizations, and locations. The B-NER dataset [2] and FIRE/ICON evaluation campaigns have provided additional Bengali NER benchmarks, though with inconsistent annotation schemes and limited public availability. Prior work on Bengali NER has largely focused on feature engineering for CRFs [8] or direct fine-tuning of multilingual transformers, without systematic evaluation under controlled data scarcity.

D. Low-Resource NER

The MultiCoNER and MultiCoNER2 shared tasks [9] highlighted the difficulty of NER under low-resource and cross-lingual conditions, motivating data-efficient approaches. Recent work combining multilingual pretrained models with structured decoding — such as MuRIL+CRF [10] — demonstrates that structured output layers improve transformer performance even when the encoder is pretrained. Cross-lingual transfer for low-resource NER [11] further shows that models with explicit linguistic inductive biases generalize better when target-language data is scarce. Our work differs in that we study models in isolation to characterize their individual data-efficiency profiles, rather than combining them in hybrid architectures.

III. MODEL FORMULATION

A. Task Definition

Let $x = (x_1, x_2, \dots, x_T)$ denote a sentence of T Bengali tokens and $y = (y_1, y_2, \dots, y_T)$ the corresponding BIO label sequence over the tag set

$$\mathcal{Y} = \{O, B\text{-}PER, I\text{-}PER, B\text{-}ORG, I\text{-}ORG, B\text{-}LOC, I\text{-}LOC\}.$$

The NER task is to find

$$\hat{y} = \arg \max_y P(y | x).$$

B. Basic CRF

The Basic CRF defines:

$$P(y | x) = \frac{1}{Z(x)} \exp \left(\sum_{t=1}^T \sum_k \lambda_k f_k(y_t, y_{t-1}, x, t) \right)$$

where f_k are feature functions over the current label, previous label, and observation, λ_k are learned weights, and $Z(x)$ is the partition function computed via the forward algorithm.

Training minimizes the negative log-likelihood with L2 regularization using L-BFGS. Feature functions for the Basic CRF include word identity, lowercased form, character prefix/suffix n-grams of length 2 and 3, binary flags (`is_digit`, `contains_hyphen`), and a one-word context window.

C. Latent Morphology CRF

Bengali is agglutinative: surface token forms vary extensively through suffixation, making word-identity features sparse across training and test distributions. To address this, we introduce a discrete latent variable

$$z_t \in \{0, 1, 2, 3, 4\}$$

representing the suffix length of token x_t , inducing the segmentation

$$(x_t^{\text{stem}}, x_t^{\text{suffix}}).$$

The joint model over labels and latent segmentations is:

$$P(y, z | x) = \frac{1}{Z(x)} \prod_{t=1}^T \phi(y_t, y_{t-1}, x_t, z_t) \cdot P(z_t | x_t)$$

where ϕ is the CRF potential and $P(z_t | x_t)$ is the morphological prior, parameterized as a softmax over learned log-weights indexed by the candidate suffix string.

The marginal likelihood used for training is:

$$P(y | x) = \sum_z P(y, z | x)$$

D. EM Training Procedure

Exact EM over the joint model is intractable due to the interaction between the CRF partition function and the latent variable summation. We therefore adopt an approximate EM procedure.

1) *E-step*: Given the current CRF parameters $\theta^{(k)}$ and morphological prior $P^{(k)}(z \mid x)$, compute for each token t the NER confidence score:

$$\text{conf}_t = P^{(k)}(y_t^* \mid x; \theta^{(k)})$$

obtained from the CRF marginal distribution via the forward-backward algorithm, where y_t^* is the true label.

Soft suffix counts are accumulated as:

$$\tilde{c}(\text{sfx}) += P^{(k)}(z_t = |\text{sfx}| \mid x_t) \cdot \text{conf}_t$$

2) *M-step (CRF)*: Retrain the CRF using features extracted from the current MAP segmentations inferred from $P^{(k)}(z \mid x)$.

3) *M-step (Segmenter)*: Update log-suffix weights by normalizing soft counts within each suffix-length bucket, with Laplace smoothing ($\alpha = 0.1$):

$$\log P^{(k+1)}(\text{sfx}) = \log \left(\frac{\tilde{c}(\text{sfx}) + \alpha}{\sum_{\text{sfx}': |\text{sfx}'|=|\text{sfx}|} (\tilde{c}(\text{sfx}') + \alpha)} \right)$$

Within-bucket normalization prevents the empty suffix (length 0, present in every token) from dominating via raw frequency. The default log-weight for unseen suffixes is set to -2.0 , corresponding to a background probability of

$$\exp(-2) \approx 0.135.$$

Three EM iterations are run per experiment.

E. Feature Groups for Ablation

The Latent Morph CRF uses five feature groups, each removable independently:

- **Morphology**: MAP stem, MAP suffix, top-1 and top-2 suffix candidates, top-1 suffix probability, and morphological entropy

$$H = - \sum_z P(z \mid x_t) \log P(z \mid x_t).$$

- **Character N-grams**: prefix-2, prefix-3, suffix-2, suffix-3 character n-grams.
- **Context**: previous and next word form, MAP suffix, and suffix-3 of neighboring tokens; BOS/EOS boundary flags.
- **Word Identity**: surface form, lowercased form, and word length.
- **Miscellaneous (never ablated)**: bias term, `is_digit`, `contains_hyphen`, `contains_number`.

F. Neural Baselines

1) *BiLSTM*: Tokens are embedded into 128-dimensional vectors using a learned embedding table initialized randomly. A bidirectional LSTM with 128 hidden units per direction produces contextual representations, which are projected by a linear layer to $|\mathcal{Y}|$ logits and trained using cross-entropy loss with Adam ($\text{lr} = 10^{-3}$) for 3 epochs and batch size 32. Sentences are padded or truncated to 50 tokens.

2) *mBERT*: We fine-tune `bert-base-multilingual-cased` [1], a 12-layer transformer pretrained on 104-language Wikipedia, for token classification.

The subword sequence is obtained using the WordPiece tokenizer. NER labels are assigned to the first subword of each original token, while remaining subwords are masked (-100) in the loss.

Training uses AdamW ($\text{lr} = 2 \times 10^{-5}$, weight decay = 0.01) with linear warmup over 10% of total steps, for 3 epochs, batch size 16, and maximum sequence length 128.

IV. EXPERIMENTAL SETUP

A. Dataset

We use the WikiANN Bengali split [?] accessed via the HuggingFace datasets library using `load_dataset("wikiann", "bn")`.

The dataset contains Wikipedia-derived sentences annotated with BIO tags for three entity types: PER (person), ORG (organization), and LOC (location). The label distribution is class-imbalanced, with `O` tokens comprising approximately 75–80% of all tokens. This imbalance motivates the use of F1-score rather than accuracy as the primary evaluation metric.

B. Low-Resource Simulation

To simulate varying degrees of data availability, we randomly sample six subsets of the training split at proportions

$$p \in \{0.01, 0.05, 0.10, 0.30, 0.50, 1.00\},$$

corresponding to 1%, 5%, 10%, 30%, 50%, and 100% of the training data.

All experiments share a fixed test set. A random seed of 42 is set globally across NumPy, Python `random`, and PyTorch to ensure reproducibility. Subsets are sampled without stratification. The 10% subset is preserved and reused identically for the ablation study.

C. Evaluation Metrics

All models are evaluated using span-level F1-score, precision, recall, and token accuracy via the `seqeval` library.

Span-level evaluation counts an entity as correct only if both its complete span and entity type match exactly; partial boundary matches receive no credit. F1-score is used as the primary evaluation metric, while accuracy is reported as a secondary diagnostic measure.

D. Implementation Details

CRF models use `sklearn-crfsuite` with L-BFGS optimization,

$$c_1 = c_2 = 0.1,$$

maximum 100 iterations, and `all_possible_transitions=True`.

The BiLSTM baseline is implemented in PyTorch. mBERT is loaded and fine-tuned using the HuggingFace transformers library.

All experiments are conducted using a single random seed, and no ensembling is performed. Training is conducted on CPU for CRF models and GPU (CUDA when available) for neural models.

V. RESULTS

A. Main Results

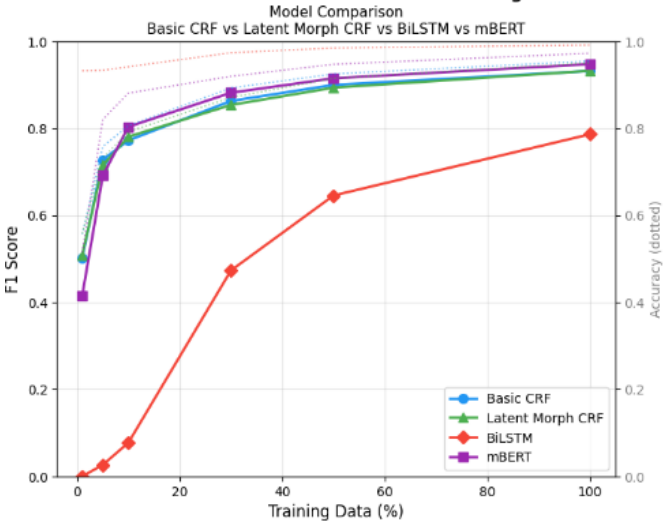


Fig. 1. Performance comparison of all models across different training data sizes.

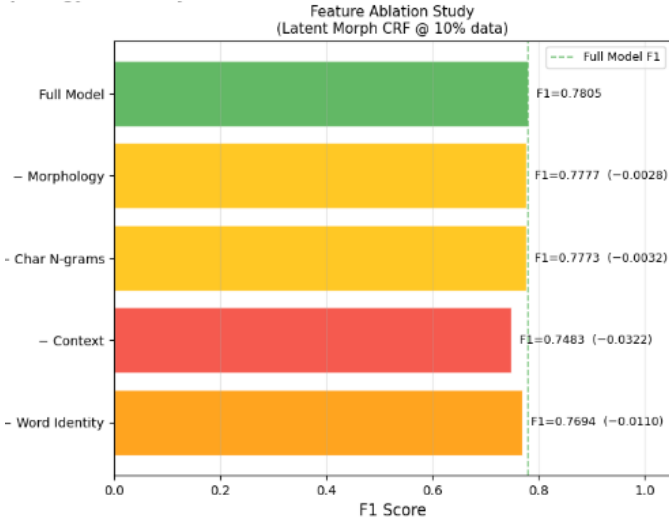


Fig. 2. Feature ablation study for the Latent Morph CRF at 10% training data.

Table I reports F1-score and accuracy for all four models across six training data sizes.

Three distinct regimes emerge from Table I.

Regime 1 — Extreme Scarcity (1–5%): CRF models dominate under severe data scarcity. At 1% training data, the BiLSTM completely fails to learn meaningful entity representations ($F1 = 0.000$), predicting the majority $\circ$ label for

nearly all tokens. This behavior yields deceptively high token accuracy while providing no practical NER utility.

mBERT, despite multilingual pretraining, also underperforms the CRF models at 1% data ($F1 = 0.416$), suggesting that approximately 120 training sentences provide insufficient fine-tuning signal for adapting multilingual subword representations to Bengali entity boundaries.

The Latent Morph CRF achieves the best result at 1% data ($F1 = 0.509$), while the Basic CRF leads at 5% ($F1 = 0.728$). These results demonstrate that explicit linguistic feature engineering provides strong inductive bias in extremely low-resource settings.

Regime 2 — Crossover (10%): At 10% training data, mBERT ($F1 = 0.803$) surpasses the Latent Morph CRF ($F1 = 0.781$) for the first time. This crossover occurs earlier than prior low-resource NER literature commonly reports, where neural advantages typically emerge at substantially larger data scales.

One possible explanation is that mBERT was pretrained on multilingual Wikipedia corpora, closely matching the WikiANN domain and therefore reducing the domain adaptation burden during fine-tuning.

Regime 3 — Data Sufficiency (30–100%): As training data increases, mBERT consolidates its advantage, reaching $F1 = 0.947$ at full data. CRF models continue improving steadily, with the Latent Morph CRF reaching $F1 = 0.932$, though the performance gap widens with increasing data availability.

The BiLSTM recovers from its early collapse but remains consistently weaker than all CRF variants. Even at 100% data, its F1-score of 0.786 remains substantially below both CRF models, indicating that random initialization without pretraining is a major limitation for Bengali NER.

B. Latent Morph CRF vs. Basic CRF

The Latent Morph CRF and Basic CRF remain closely matched across most training scales. The Latent variant leads slightly at 1% data (+0.007 F1) and again at 100% data (+0.003 F1).

Although the F1 improvements are modest, the Latent Morph CRF provides additional interpretability through its learned suffix distribution. At 100% training data, the highest-probability suffixes include

These suffixes correspond to linguistically meaningful Bengali nominal case endings and verb inflections, suggesting that the EM procedure successfully captures coherent morphological regularities.

C. Ablation Study

Table II presents the feature ablation results for the Latent Morph CRF using the 10% training subset.

Context features are by far the most important component, producing an F1 reduction of 0.032 when removed — nearly ten times larger than any other feature group. This indicates that neighboring word information provides the dominant signal for resolving entity boundaries at moderate data scales.

TABLE I
F1-SCORE BY MODEL AND TRAINING DATA SIZE

Data %	Basic CRF	Latent CRF	BiLSTM	mBERT	Best Model
1%	0.5021	0.5091	0.0000	0.4156	Latent CRF
5%	0.7277	0.7150	0.0261	0.6911	Basic CRF
10%	0.7721	0.7805	0.0773	0.8034	mBERT
30%	0.8627	0.8536	0.4731	0.8816	mBERT
50%	0.8996	0.8933	0.6454	0.9148	mBERT
100%	0.9317	0.9320	0.7862	0.9474	mBERT

TABLE II
FEATURE ABLATION RESULTS FOR LATENT MORPH CRF AT 10% DATA

Configuration	F1	Precision	Recall	F1 Drop
Full Model	0.7805	0.7823	0.7787	—
– Context	0.7483	0.7585	0.7383	−0.0322
– Word Identity	0.7694	0.7712	0.7677	−0.0111
– Char N-grams	0.7773	0.7806	0.7741	−0.0032
– Morphology	0.7777	0.7879	0.7677	−0.0028

Word identity is the second most influential feature group, contributing direct lexical memorization of entity forms observed during training.

Character n-grams and morphological features each provide modest but consistent improvements, with F1 reductions of approximately 0.003 when removed. These results suggest that subword information functions primarily as complementary rather than primary evidence.

The relatively small contribution of morphology at 10% data warrants interpretation. One plausible explanation is that contextual features already capture sufficient local disambiguation information, leaving limited residual uncertainty for morphology to resolve. Under more extreme scarcity conditions (e.g., 1% data), morphological features may play a larger role, though evaluating this hypothesis is left for future work.

VI. DISCUSSION

A. When Does Structure Help?

Our results provide empirical support for a central prediction of structured probabilistic modeling: explicit inductive biases are most valuable when training data is insufficient for neural models to learn those biases directly from statistical regularities.

Below 10% of the WikiANN Bengali training split, CRF-based models consistently outperform neural alternatives. The CRF’s hand-engineered feature functions — including contextual windows, character n-grams, and morphological patterns — provide reliable signal under extreme data scarcity, where neural models fail to generalize effectively.

This low-resource regime is not merely theoretical. Many practical Bengali NLP applications, including regional dialect processing, specialized domains such as legal or medical text, and crisis-response systems, operate with annotation budgets well below this threshold.

B. The BiLSTM Failure Mode

The complete collapse of the BiLSTM at 1% data (F1 = 0.000, Accuracy = 0.932) is the most striking result of this study.

The model learns to predict the majority $\circ$ label for nearly all tokens — an optimal strategy under cross-entropy loss given severe class imbalance, but one that renders the model effectively useless for NER.

This failure occurs because randomly initialized word embeddings cannot form meaningful semantic representations from approximately 120 training sentences. Unlike mBERT, the BiLSTM receives no pretraining signal to compensate for limited supervision.

The deceptively high token accuracy is therefore not evidence of successful learning; rather, it reflects a degenerate majority-class prediction strategy. This observation further validates the use of span-level F1-score as the primary evaluation metric, since accuracy alone would entirely obscure this failure mode.

C. The Crossover at 10%

The finding that mBERT overtakes CRF models at approximately 10% training data — earlier than the 30–50% range often cited in low-resource NER literature — likely reflects characteristics specific to this experimental setting.

First, the domain alignment between mBERT’s pretraining corpus (multilingual Wikipedia) and WikiANN (also Wikipedia-derived) is unusually favorable. As a result, mBERT’s representations are already partially adapted to the stylistic and lexical properties of the evaluation corpus.

Second, Bengali is explicitly included in mBERT pretraining. Consequently, the model possesses direct Bengali subword coverage rather than relying on transliteration or indirect cross-lingual transfer.

In genuinely out-of-domain or zero-shot settings, the crossover threshold would likely shift toward substantially higher data percentages, extending the regime in which CRF-based models remain competitive.

D. Practical Recommendations

Based on these findings, we propose the following guidelines for practitioners developing Bengali NER systems under varying resource constraints:

- **Less than 10% labeled data available:** Use the Latent Morph CRF or Basic CRF. Both models train efficiently

on CPU, provide interpretable predictions, and consistently outperform neural baselines in extremely low-resource settings.

The Latent Morph CRF is preferable when morphological interpretability is important, while the Basic CRF offers a simpler implementation with comparable performance.

- **At least 10% labeled data available:** Fine-tune mBERT. Its performance advantage becomes consistent beyond the crossover point and increases steadily with additional data.

We recommend AdamW optimization with linear warmup and a maximum sequence length of 128 to accommodate Bengali subword tokenization effectively.

- **Avoid BiLSTM without pretrained embeddings:** Even at 100% training data, the BiLSTM achieves only $F1 = 0.786$, remaining below all CRF variants.

This result indicates that random initialization is not merely a low-data issue but a persistent limitation for Bengali NER without external pretrained representations.

VII. CONCLUSION

This paper presented a systematic comparison between structured probabilistic models and neural baselines for Bengali Named Entity Recognition under controlled low-resource conditions.

We introduced a Latent Morphology CRF that jointly learns Bengali suffix distributions and NER labels through an Expectation-Maximization (EM) procedure. The proposed model was evaluated alongside a Basic CRF, BiLSTM, and multilingual BERT (mBERT) across six training data fractions ranging from 1% to 100% of the WikiANN Bengali corpus.

Our experiments yield four primary findings:

- 1) CRF-based models significantly outperform neural baselines under extreme data scarcity (1–5% training data). In particular, the BiLSTM collapses entirely at 1% data due to the absence of pretrained representations.
- 2) The crossover point at which mBERT surpasses CRF models occurs at approximately 10% training data, earlier than commonly reported in prior low-resource NER literature. This result is likely driven by the strong domain alignment between mBERT pretraining and the WikiANN corpus.
- 3) Contextual features are the dominant signal source within the Latent Morph CRF at moderate data scales, while morphological features contribute smaller but consistent supplementary improvements.
- 4) BiLSTM without pretrained embeddings underperforms across all training scales and is therefore not recommended for Bengali NER.

These findings contribute to a clearer understanding of the data-efficiency tradeoffs between structured probabilistic models and pretrained neural architectures, while also providing concrete deployment guidelines for Bengali NLP practitioners.

Future work will extend this analysis to out-of-domain Bengali corpora, investigate hybrid mBERT+CRF architectures, and apply the latent morphological segmentation framework

to other morphologically rich low-resource languages such as Turkish and Swahili.

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